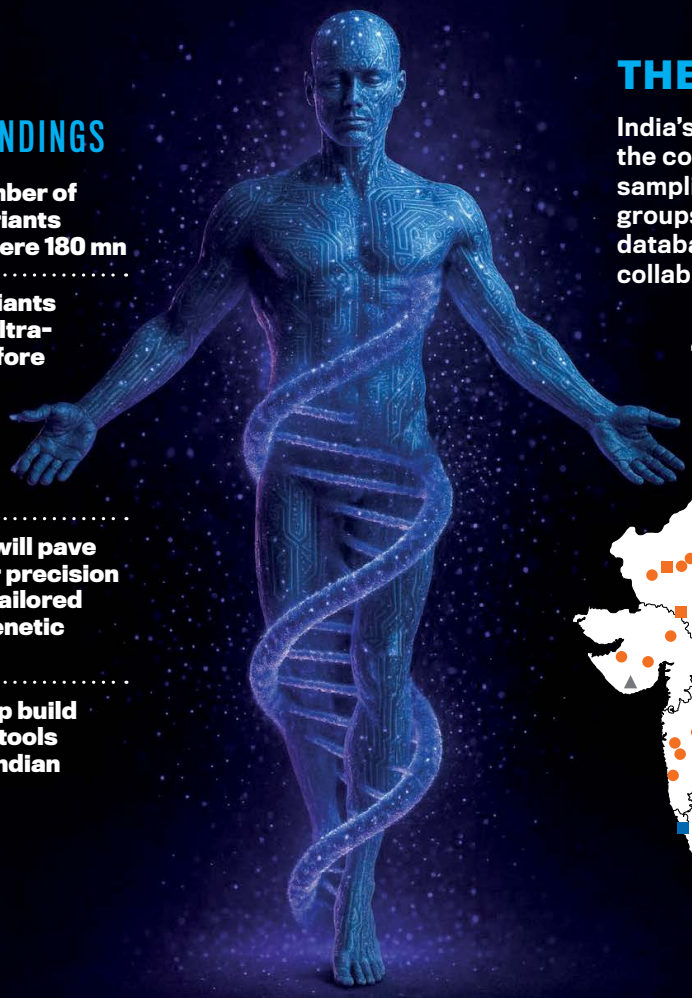
EARLY FINDINGS

- Total number of genetic variants obtained were 180 mn
- Most variants (65%) are ultra-rare, therefore important to understand complex diseases
- Dataset will pave the way for precision medicine, tailored to Indian genetic profiles
- It will help build diagnostic tools suited for Indian population



THE GENOME BLUEPRINT

India's largest study to identify the country's genetic diversity by sampling its vast number of ethnic groups creates a first-ever genome database through a nationwide collaboration of scientific institutes

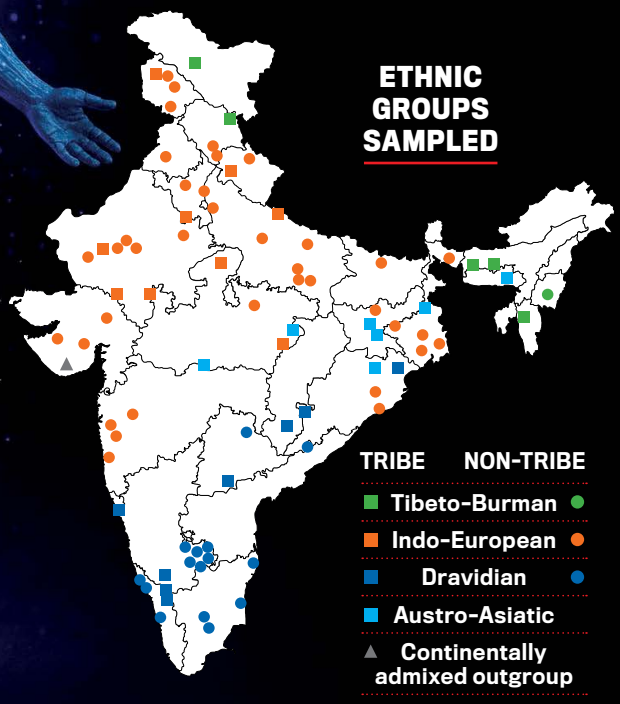


Illustration by **NILANJAN DAS/ AI**



"ONCE THE INFORMATION FROM THE GENOME SEQUENCE OF 9,772 INDIVIDUALS IS EXTRACTED, WE WILL HAVE VERY COMPREHENSIVE INFORMATION ABOUT THE INDIAN POPULATION, ITS HISTORY AS WELL AS MEDICAL IMPLICATIONS"

DR KUMARASAMY THANGARAJ
Joint National Coordinator, GenomeIndia

enhanced diagnostics. The former means that clinicians can tailor treatment and prescribe drugs based on genetic profiles, taking into account an individual's response, particularly an adverse reaction, to a drug. In the area of diagnostics, the GenomeIndia project aims to design a genotype array for the Indian population which will improve on existing arrays. Unlike whole genome sequencing, a genotyping array is a genetic test that takes readouts from certain markers in the genome and analyses it in conjunction with a reference panel—this makes it more affordable than whole genome sequencing. Currently, given that the global genomic landscape is predominantly Euro-centric, models built on other populations have limitations.

"This Indian genetic dataset will help build better models to predict a person's risk of developing diseases like type 2 diabetes or Alzheimer's, which are polygenic in nature, meaning an individual's susceptibility is determined by many genes and influences from environmental factors," explains Bratati Kahali, one of the principal investigators based in the Centre for Brain Research at the Indian Institute of Science, Bengaluru, which is the project's coordinating centre. "Right now, most of these models are based on data from European populations, so they don't work as well for Indians. The GenomeIndia data reveals genetic patterns specific to the Indian population, which can improve the accuracy of these models for Indians and possibly other South Asians. Overall, this research will help us better understand how genetics plays a role in different diseases."

Having accomplished the arduous task of setting up an information-rich biobank, researchers are now bracing to mine the data for insights that can bring precision to India's healthcare. ■